FINANCIAL MANAGEMENT
Suggested Answers
July-August 2023

Answer to Question# 1 (a):

Briefing Note for Bijli Aluminum Limited's Management:

Subject: Explanation of Portfolio Theory, Systematic Risk, Unsystematic Risk, and Coefficient of Correlation

Dear Management,

I am writing to provide you with an explanation of key financial concepts that are relevant to your current strategy of diversification into the cement industry, as mentioned in the executive summary received from the management consultants. Understanding these concepts will help you make informed decisions about your investment strategy.

i) Portfolio Theory: Portfolio theory, also known as Modern Portfolio Theory (MPT), is an investment concept developed by Harry Markowitz. It suggests that investors can optimize their portfolio by selecting a combination of assets (stocks, bonds, etc.) that maximizes expected returns for a given level of risk or minimizes risk for a given level of expected returns. The core idea behind portfolio theory is that combining different assets that are not perfectly correlated can reduce overall risk while maintaining or even improving returns.

ii) Systematic Risk and Unsystematic Risk:

- **Systematic Risk (Market Risk):** This is the risk that is inherent in the overall market or economy and affects all investments. It's also known as non-diversifiable risk because it cannot be eliminated through diversification. Factors like interest rate changes, economic recessions, political events, and market volatility contribute to systematic risk. It's the risk that investors are rewarded for taking on over the long term.
- **Unsystematic Risk (Specific Risk):** This is the risk that is unique to a specific company or industry. It can be eliminated or significantly reduced through diversification. Examples of unsystematic risk include company-specific events, management changes, technological advancements, and industry-specific trends. Investors are not rewarded for taking on unsystematic risk because it can be diversified away.

iii) Coefficient of Correlation: The coefficient of correlation measures the degree of relationship between two sets of data. In the context of your investment strategy, it indicates the extent to which the prices or returns of two industries (aluminum and cement) move together. The coefficient of correlation can have values between -1 and +1:

- **Positive Correlation (+1):** A positive correlation indicates that the two industries tend to move in the same direction. A coefficient of +1 suggests a perfect positive correlation, meaning that when one industry's prices or returns go up, the other industry's prices or returns also go up.

- **Negative Correlation (-1):** A negative correlation indicates that the two industries tend to move in opposite directions. A coefficient of -1 suggests a perfect negative correlation, meaning that when one industry's prices or returns go up, the other industry's prices or returns go down.
- **Zero Correlation (0):** A zero correlation means that there is no clear relationship between the two industries' prices or returns. Changes in one industry do not predict changes in the other.

In your case, the coefficient of correlation between the aluminum and cement industries is +0.8. This indicates a strong positive correlation, suggesting that the two industries tend to move together. While portfolio theory advocates diversification to reduce risk, the high positive correlation between aluminum and cement raises questions about how much risk can actually be diversified away.

In conclusion, portfolio theory encourages diversification to manage risk, with the goal of reducing unsystematic risk. However, the strong positive correlation between the aluminum and cement industries may limit the extent to which risk can be diversified. It's important to carefully evaluate the potential benefits and risks of diversifying into the cement industry considering this correlation.

Please feel free to reach out if you need further clarification or assistance.

Sincerely,

[Your Name]

Finance Manager, Bijli Aluminum Limited

Answer to Question# 1 (b) (i):

i) 'In view of the fact that the market is efficient in the semi-strong form, financial information released by companies is of no value to investors, because the information is already included in share prices before it is released':

This statement refers to the semi-strong form of market efficiency and its implications for the value of financial information to investors.

Semi-Strong Form of Market Efficiency: In the context of market efficiency, the semi-strong form suggests that all publicly available information is already reflected in stock prices. This includes not only past price and trading volume data (weak form), but also all publicly available information such as financial reports, news, and announcements (semi-strong form).

Comment: The statement is correct in its assertion that, according to the semi-strong form of market efficiency, publicly available financial information is quickly and accurately incorporated into share prices. Therefore, investors cannot consistently earn abnormal returns by trading on publicly available information alone. However, it's

important to note that private or insider information is not included in this argument, as its utilization can lead to an advantage in the market.

ii) 'If an investor holds shares in about 20 different companies all of the risk is eliminated and the portfolio will give a return equal to the risk-free rate':

This statement involves the concept of diversification and its impact on risk and returns in an investment portfolio.

Diversification: Diversification is the strategy of investing in a variety of assets or securities to reduce the overall risk of the portfolio. It's based on the idea that by holding a mix of investments that are not perfectly correlated, the potential for large losses due to the poor performance of a single investment is minimized.

Risk-Free Rate: The risk-free rate is the theoretical rate of return of an investment with zero risk, usually represented by the yield on government bonds.

Comment: While diversification does reduce risk, it doesn't eliminate it entirely. Holding shares in different companies can help mitigate company-specific risk (unsystematic risk), but systematic risks like market fluctuations and economic changes can still impact the overall portfolio. Additionally, the assertion that the portfolio will give a return equal to the risk-free rate is not accurate. The return of a diversified portfolio will generally exceed the risk-free rate because investors expect to be compensated for taking on additional risk.

iii) 'A graph of the daily price of a share looks similar to that which would be obtained by plotting a series of cumulative random numbers. This shows clearly that share prices move randomly at the whim of investors, indicating that the market is not price efficient':

This statement alludes to the idea that stock prices exhibit randomness and challenges the concept of market efficiency.

Random Walk Hypothesis: The random walk hypothesis posits that stock prices move in a sequence of independent random steps, with future price changes not dependent on past price movements. This suggests that predicting stock price movements based on historical data is difficult.

Market Efficiency: The concept of market efficiency asserts that stock prices reflect all available information, making it difficult for investors to consistently beat the market by using publicly available information.

Comment: The observation of a price graph resembling cumulative random numbers is consistent with the random walk hypothesis. However, this does not necessarily negate market efficiency. Market efficiency doesn't imply that stock prices are completely predictable in the short term; rather, it suggests that prices fully reflect available information. Over the long term, market efficiency still holds, and investors should not be able to consistently earn abnormal profits by trading on historical price patterns alone. The notion of market efficiency does not preclude short-term price fluctuations driven by investor sentiment or other factors that can appear random.

Answer to Question# 2 (a):

Calculation of weighted average cost of capital

Definition

k = Weighted average cost of capital
ke = Cost of equity capital
kd = Cost of debenture capital
kL = Cost of bank loan
E = Total ex-dividend market value of debt
D = Total ex-interest market value of loan
L = Total value of outstanding bank loan

i) Calculation of ke

Assuming an underlying dividend growth 'g' per anumn, the average growth rate between 2018 and 2022 is given by

$(1+g)^4 = 13.6 \div 10$
$= 1.36$
$(1+g) = \sqrt[4]{1.36}$
$= 1.0799$
g 8%

Assuming that shareholders take past dividend growth as a reasonable approximation to future growth, then using the dividend growth model,

$k_e = (D_0 (1 + g)) \div P_0 + 0.08$
$= (13.6 \times 1.08) \div 135 + 0.08$
$= 18.88\%$

ii) Calculation of kd

kd is the discount rate which equates the present value of future income (Tk. 8 per anumn) and redemption (Tk. 100) to the current market price (Tk. 82.50)

Time	Flow	DF @ 10%	PV	DF @ 15%	PV
1-4	8.00	3.170	25.36	2.855	22.84
4	100.00	0.683	68.30	0.572	57.20
0	(82.50)	1.000	(82.50)	1.000	(82.50)
			11		(2)

By linear interpolation

kd = 10 + (11.16 ÷ (11.16+2.46)) x (15-10)
= 14%

iii) Calculation of cost of bank loan

The current cost of debt is taken as the best estimate of the future cost of debt i.e., 16.5%

iv) Calculation of weighted average cost of capital

k = (ke E + kd + kL L) ÷ (E + D + L)
= (4.05 x 18.88% + (0.66x14%) + (0.9x16.5%)) ÷ (4.05+0.66+0.9)

= 18%

Answer to Question# 2 (b):

Assumption underlying the use of k as a discount rate

The weighted average cost of capital should only be used as the target discount rate for appraising investment opportunities whose acceptance will not alter the weighted average cost of capital

Since the cost of any type of capital can be regarded as a function of a risk free rate and a risk premium, this implies that k should not be used to evaluate opportunities which have significantly different risk characteristics from the average risk borne by the company prior to acceptance of the project.

In this context it is useful to separate the total risk of the company into business risk and financial risk. The business risk is the risk inherent in the nature of the company's operations. The financial risk is a function of a company's gearing. For a project to be evaluated using k, its acceptance must not alter the company's overall business risk nor must it alter the financial risk. Therefore, it must be of a similar nature to existing projects and it must be financed in such a way that the gearing ratio in unchanged and hence the financial risk is unaltered. In practice k may also be used to evaluate small or marginal projects whose acceptance is unlikely to alter overall corporate risk.

The situation outlined in the question is such that the project being considered could hardly be thought of as marginal. The cost of the project (Tk. 2.5 million) is approximately half of the existing market value of the company. In these circumstances k could only be used as a target discount rate if its business risk were the same as that of existing projects and it were to be financed in the same way as existing projects. This is unlikely to be the case, and as a generalization it is probably unwise to use the existing k to evaluate such a major investment opportunity.

Answer to Question# 2 (c):

Practical problems in estimating k for a large Bangladeshi listed company

The problems of estimating k can be considered under four headings:

i) The validity of the model
The estimation of k and, in particular ke is based on the assumption that the value of an ordinary share is the discounted present value of the future dividend stream. It is possible that the market is using a different method of valuation, eg based on an earnings multiple; if this is the case, validity of the ke calculation is undermined.

ii) The estimation of the variables in the model
Here again the major problems is in the estimation of ke. To calculate ke it is necessary to know or estimate Do, Po, g. Of these only Do can be determined with confidence. Po is the ex-dividend value of the share but, for the model to give an accurate estimate of ke, the market value must be in equilibrium. In practice the market value of an ordinary share may fluctuate daily and it may be extremely difficult to estimate its equilibrium value.

If establishing Po is a problem, then estimating g is harder still. Two approaches are frequently used, one based on past dividend growth and one using the Gordon growth model. In using past growth rates an average must be calculated and there is always the danger that the average is misleading. In the question the average dividend growth of ABC Limited over the past four years has been 8%, although in the most recent year dividends have not grown at all. To what extent is the average growth a meaningful figure? Furthermore, it must be remembered that the growth factor in the dividend valuation model is anticipated

future growth. Even where the average of past growth rates is meaningful, the use of the average assumes that past growth rates will be sustained in the future.

The earnings retention model is based on the belief that future dividend growth depends on the volume of earnings retained in the business and the rate of return earned on retained earnings.

The estimate of g is given by

g	=	br
Where, b	=	the current retention rate
r	=	the current return on capital employed

Even accepting the naivety of the model and its limitation to pure equity companies, it will only give a useful approximation to g where the retention rate and the rate of return are stable.

iii) Fluctuation in the value of the variable through time
Once the basic rationale for a discounting approach to investment appraisal has been accepted (i.e., both the magnitude and timing of cash flows are relevant), it is incumbent upon the company to establish an appropriate discount rate. The WACC model is a reasonable start to determining the appropriate rate, notwithstanding the obvious difficulties in its application. In practice, a check on the calculations can be made by comparing the result with the WACC of similar companies and with the risk-free rate (the return on government securities)

iv) Problems associated with different sources of finance
In part (a) a calculation was made of the WACC. ABC limited is financed entirely by debt and ordinary share capital, and the procedure for calculating the WACC is straightforward. However, in particular, many firms use other sources of finance, which make the computation of WACC more difficult.

Specifically, a company may have unlisted preference shares which present problems of estimating their market value; the estimation of true cost of convertible loan stock is notoriously difficult as the calculation requires an assumption about whether or not the option to convert will be exercised; where a company has substantial foreign currency loans, fluctuations in the exchange rate add a new dimension to the estimate of their cost; and, finally. substantial 'off balance sheet' finance in the form of lease agreements creates further difficulties.

Answer to Question# 3:

Calculate the effects of each of the four approaches

i) Convert at spot in three months' time
If exchange rate is Tk. 1/€1.50
€5 m /1.50 = Tk 3,333,333

If exchange rate is Tk. 1/€1.70
€5 m /1.70 = Tk 2,941,176

ii) Use the forward market
3 - month forward sell rate = 1.5625 – 0.0015 = 1,561

€5 m /1.561 = Tk 3,203,075

iii) Exchange rate in three months: Tk. 1/€1.50
In this case the company will not exercise the option but will convert at spot
€5 m /1.50 = Tk. 3,333,333
Net outcome after premium Tk 3,208,333

Exchange rate in three months: Tk. 1/€1.70
In this case the company will exercise the option and convert at the strike price of €1. 561
€5 m /1.561 = Tk. 3,203,075

Net outcome after premium Tk 3,078,075

iv) Money market cover

Borrow = €5 m /(1+(0.032/3))
= € 4,960,317

Convert at spot
= €4,960,317/ 1.5625
Tk. 3,174,603

Invest for three months
= Tk. 3,174,603x (1+0.015)
= Tk. 3,127,603

Summary:
Cash receipt in three months' time (in Tk.)

		Spot rate in three months	
		Tk. 1/€1.50	Tk. 1/€1.50
(i)	Conversion at spot	3,333,333	2,941,176
(ii)	Using the forward market	3,203,075	3,203,075
(iii)	Buying a €5 m put option	3,208,333	3,078,075
(iv)	Borrowing € and investing Tk.	3,127,603	3,127,603

Answer to Question# 4 (a):

The beta of a portfolio asset is the weighted average of the betas of the individual assets in the portfolio, with the weight on each asset being that asset's share in the portfolio. Specifically,

$$\beta_p = \sum_{j=1}^{n} w_j X \beta_j and \sum_{j=1}^{n} w_j = 1 \rightarrow \beta_p = (w_A \times \beta_A) + (w_B \times \beta_B)\,(w_C \times \beta_C) + (w_D \times \beta_D)$$

Asset	Cost	Weight (W_1)	Beta (β/)	$W_j \times \beta_1$
Asset	20,000	20%	0.80	0.1600
B	35,000	35%	0.95	0.3325
C	30,000	30%	1.50	0.4500
D	15,000	15%	1.25	0.1875
Total=	100,000	100%	Sum = β_p =	1.1300

Answer to Question# 4 (b):

$$\text{Return} = \frac{[(ending\ value - initial\ value) + cash\ distribution}{initial\ value}$$

$$r_A = \frac{(Tk.20{,}000 - Tk.20{,}000) + Tk.1{,}600}{Tk.20{,}000} = \frac{Tk.1{,}600}{Tk.20{,}000} = 8\%$$

$$r_C = \frac{(Tk.34{,}500 - Tk.30{,}000) + Tk.0}{Tk.30{,}000} = \frac{Tk.4{,}500}{Tk.30{,}000} = 15\%$$

$$r_B = \frac{(Tk.36{,}000 - Tk.35{,}000) + Tk.1{,}400}{Tk.35{,}000} = \frac{Tk.2{,}400}{Tk.35{,}000} = 6.86\%$$

$$r_D = \frac{(Tk.16{,}500 - Tk.15{,}000) + Tk.375}{Tk.15{,}000} = \frac{Tk.1{,}875}{Tk.15{,}000} = 12.5\%$$

Answer to Question# 4 (c):

Total value of the portfolio today is Tk. 107,500, and total annual income from Assets A-D during the year was Tk. 3,375, so portfolio return is given by:

$$r_P = \frac{(Tk.107{,}000 - Tk.100{,}000) + Tk.3{,}375}{Tk.100{,}000} = \frac{Tk.10{,}375}{Tk.100{,}000} = 10.375\%$$

Answer to Question# 4 (d):

Using the CAPM equation, $r_j = R_f + [\beta_j \times (r_m - R_f)]$, where r_j is the required return on asset j.

R_F the risk-free rate (given as 4%), β_j the beta on asset$_j$ and j, r_w the return on the market portfolio return (given as 10%), solve for required return on each project:

$r_A = 4\% + [0.80 \times (10\% - 4\%)] = 8.8\%$
$r_B = 4\% + [1.50 \times (10\% - 4\%)] = 13.0\%$
$r_C = 4\% + [0.95 \times (10\% - 4\%)] = 9.7\%$
$r_D = 4\% + [1.25 \times (10\% - 4\%)] = 11.5\%$

Answer to Question# 4 (e):

Asset	Actual Return	CAPM Return	Over- or Underperformed CAPM
A	8.00%	8.80%	Under
B	6.86%	9.70%	Under
C	15.00%	13.00%	Over
D	12.50%	11.50%	Over

Security analysts typically use statistical techniques to estimate an asset's beta by obtaining a line of best fit through historical asset and market returns. The slope of this line is beta. Data points-that is, actual returns on the asset and market for a given period—will be randomly scattered around the line no matter how well it "fits" the data. The point here is asset betas are estimates. The CAPM return is, in a sense, a forecast and even good forecasts are subject to random error. Another possibility is beta does not fully capture all no diversifiable or systemic factors that affect expected returns. Still another possibility is the firm behind the asset has changed, so the beta estimated with historical data does not reflect the asset's current beta.

Answer to Question# 5 (a):

	30 June				
	2023	**2024**	**2025**	**2026**	**2027-43**
	Y0	**Y1**	**Y2**	**Y3**	**Y4-20**
	BDT '000	**BDT '000**	**BDT '000**	**BDT '000**	**BDT '000**
Construction costs	-750,000	-750,000	-750,000		
Land clearance	-20,000				
Sales		750,000	1,250,000		
Rental income (W1)			60,000	144,000	144,000
Bad debts (W1)			-1,800	-4,320	-4,320
New staff			-4,000	-9,000	-9,000
Extra costs (W1)			-2,400	-5,760	-5,760
Tax (W2)	4,500	-92,813	-166,343	-28,107	-28,107
Green machine	0	-20,000	0	2,000	
Tax on machine (W3)	0	54	43	-277	
Total cash flows	**-765,500**	**-112,758**	**385,501**	**98,536**	**96,813**
9% factors (W4)	1	0.917	0.842	0.772	6.597
PV	-765,500	-103,400	324,592	76,070	638,700
NPV	**170,461**				

The project produces a positive NPV and so should be accepted as it will enhance shareholder wealth.

WORKINGS:

W1 Rental income, bad debt & Extra costs

	BDT '000
Rental income (Y2)	60,000
Bad debts (Y2)	1,800
Extra costs (Y2)	2,400
Rental income (Y3)	144,000
Bad debts (Y3)	4,320
Extra costs (Y3)	5,760

W2 Tax allowance and Tax

	2023 Y0 BDT '000	2024 Y1 BDT '000	2025 Y2 BDT '000	2026 Y3 BDT '000	2027-43 Y4-20 BDT '000
Construction cost for sold flats		-337,500	-562,500		
Land clearance	-20,000				
Sales		750,000	1,250,000		
Rental income (W1)			60,000	144,000	144,000
Bad debts (W1)			-1,800	-4,320	-4,320
New staff			-4,000	-9,000	-9,000
Extra costs (W1)			-2,400	-5,760	-5,760
Taxable (loss)/profit	-20,000	412,500	739,300	124,920	124,920
Tax at 22.5%	**4,500**	**-92,813**	**-166,343**	**-28,107**	**-28,107**

W3: Tax savings on capital allowance

	2024 Y1 BDT '000	2025 Y2 BDT '000	2026 Y3 BDT '000
Machine cost/WDV	1,200	960	768
WDA (20%)/Balancing allowance	-240	-192	1,232
WDV/Sale price			100
Tax saving (22.5% x WDA)	**54**	**43**	**-277**

W4: Annuity factor

9% annuity factor for Y4-Y20	Y20	9.129	OR		(8.544)
	Y4	(2.531)		x	0.772
		6.597			6.597

Answer to Question# 5 (b):

	Y1 BDT '000	Y2 BDT '000	Total BDT '000
Sales	750,000	1,250,000	
Tax	-168,750	-281,250	
Total cash flows	581,250	968,750	
9% factors	0.917	0.842	
PV	**533,257**	**815,377**	**1,348,634**

Sensitivity = 170,461 / 1,348,634 = **12.6%**

Minimum selling price = (BDT 2,500,000 – 12.6%) = **2,184,011**

Answer to Question# 5 (c):

	Y0 BDT '000	Y1 BDT '000	Y2 BDT '000
Incremental construction costs	-1,350,000	750,000	750,000
Tax on costs		-5,062.5	-5062.5
Total cash flows	-1,350,000	744,937	744,937
9% factors	1	0.917	0.842
PV	-1,350,000	683,108	627,237
	-39,655		

The NPV would decrease by BDT 39.65 million and so it is less likely that AHL's board would proceed with the development.

Answer to Question# 6 (a):

Chairman – to use the cost of preference shares would be completely wrong. It's only one element of the firm's total long-term finance and 7% is the coupon rate, not the current cost.

Finance Director and Marketing director – ordinary shares (cost of equity) should be taken into account. It would make sense to use YBF's current WACC figure for the investment appraisal if:

(1) the historical proportions of debt and equity are not to be changed
(2) the systematic business risk of the firm is not to be changed and
(3) the new finance is not project-specific.

Regarding the above, the bank borrowing will not change the gearing i.e. sufficient equity will be raised to maintain the gearing at its current level. The systematic business risk of the firm is likely to change as it's moving into a different market. The finance is not project-specific e.g. cheap government loan.

So WACC should be used which will consider all components of debt and equity

Ethical implications:
You work for YBF and are a party to confidential information which, if made public, could influence the market price of YBF's shares.

An ICAB Member should assume that all unpublished information about a prospective, current or previous client's or employer's affairs, however gained, is confidential.
That information should then:

- Be kept confidential
- Not disclosed, even inadvertently such as in a social environment
- Not be used to obtain personal advantage

Answer to Question# 6 (b):

Option 1

	2024	**2025**	**2026**
	BDT millions	**BDT millions**	**BDT millions**
Sales	528	464.64	408.88
Contribution	**396**	**348.48**	**306.6624**
Redundancy			-5
Pre-tax	**396**	**348.48**	**301.6624**
Tax @ 17%	-89.10	-78.41	-67.87
Working capital (W1)	6	5.28	38.72
Plant and equipment			55
WDAs (W2)	2.25	1.8	5.18
Total	**315.15**	**277.15**	**332.69**
Factors @ 10%	0.917	0.842	0.772
Present value	**288.99**	**233.36**	**256.84**
Total PV	**779.19**		

W-1

	2024	**2025**	**2026**
Working Capital	6	5.28	38.72

W2	**2024**	**2025**	**2026**
WDA	50	40	32

CA	10	8	6.4
Sell value			9
Balancing charge			-23
Tax	**2.25**	**1.8**	**5.18**

Option-2:

	2023
	BDT millions
Sale proceed	**775**

Option-3:

	2023	**2024**	**2025**
	BDT millions	**BDT millions**	**BDT millions**
CF	300	300	500
PV Factor @ 9%	1	0.997	0.842
	300	275	421
After tax	**772**		

The present values are:
Winding down operations 779 million
Selling to another company 775 million
MBO 772 million

To maximise shareholder wealth Padma plc should wind down operations since it produces the highest present value. However the present value relies upon a number of assumptions about sales volume, the release of working capital and the proceeds of selling plant and equipment. The present value is not sufficiently high to choose it over the other two proposals as the figures are pretty similar.

In present value terms there is little to choose between selling to another company or an MBO. Since it might be difficult to find a buyer for BSML, the preferred proposal would be for the current management team of the company to buy it.

However, if BSML goes into liquidation or having financial difficulties in future, full payment would not be received by Padma plc. So Padma plc should renegotiate with management team for higher payments in 2023 and 2024 than the previous schedule. This will increase the NPV as well as reduce the risk of default.

---The End---